

Vaishnevan Thayaseelan

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SUMMARY OF QUALIFICATIONS

Design & Analysis: 3D Printing (FDM), Mechanical Testing, SolidWorks, ANSYS Mechanical (FEA), Design Verification & Validation, AutoCAD 2D/3D, GD&T, Manufacturing Drawings (ISO), Fusion 360, Power BI, Excel, Multisim, Microsoft Office

Computing: Python (Matplotlib, NumPy, SciPy), C++, MATLAB, SQL

Interpersonal & Communication: French Proficiency, Technical Documentation, PowerPoint, Project Management, Cross-Level Reporting, G2 License

Relevant Coursework: Materials, Circuits, Electromagnetism, Engineering Graphics & CAD, Dynamics, Statics, Programming

PROJECTS

Ultimate Beyblade Design (Spinning Top with Housed Axial Fan) October 2025 – December 2025

- Manufactured, inspected, and verified components to sub-millimeter tolerances **applying GD&T** to ensure rotational stability and minimize wobbling
- 3D printed and prototyped** PLA and metallic components to construct a Beyblade with an embedded fan for axial thrust causing **168% increase in apparent weight**, determined through product testing
- Designed gear-driven mechanical machine in SolidWorks achieving a **236% RPM increase** over baseline launcher through gear ratio optimization, applying gear and mechanical advantage principles to guide design decisions
- Conducted performance testing of axial fan geometry** using aerodynamic simulation, mapping stall thresholds and tuning profile to sustain target downforce while reducing drag across the RPM range, resulting in refined airfoil profile
- Performed initial **durability testing** under combined assembly loading in ANSYS Mechanical, sizing the fan blade to **prevent collapse** from loading of the heavier exterior ring, maintaining adequate thrust and **safety under 120% of max loading**

Electric Go Kart Design January 2026 – Ongoing

- Designing a custom 48V $\approx$ 1.4 kWh lithium-ion battery pack** built by spot-welding 18650 cells in series-parallel configuration, sized to match drivetrain power and range requirements
- Designing an independently manufacturable electric go-kart chassis in SolidWorks, integrating .STEP files of off-the-shelf components to **ensure proper fit and tolerance** for respective component compartments
- Designed and prototyped cell-holding fixtures** for spot welding, iterating weld parameters and verifying joint consistency
- Identified stress concentrations** through **root-cause analysis** at mounting joints exceeding safe limits, reinforcing joint geometry through design iteration to **prevent structural failure** and ensure operator safety using ANSYS Mechanical

Battle Hams (Collaborative Toy Project) October 2025 – November 2025

- Designed and fabricated an electro-mechanical system** integrating DC motors, H-bridge motor control circuits, and a pulley-driven actuation mechanism for a two-player interactive game alongside a team of mechanical engineers
- Built and assembled** structural enclosure and test fixtures, designed with AutoCAD 2D and SolidWorks, constructed using hand tools, power tools, and an Epilog Mini 24 laser cutter, **ensuring structural integrity** and low manufacturing cost
- Prepared BOM, assembly instruction, and technical documentation** for design reviews from initial concept to final product

RELEVANT EXPERIENCE

STEM Fellowship — Eastern Regional Lead October 2024 – July 2025

- Selected as 1 of 3 Canadian Regional Leads from Canadian university students; led and mentored 8 school chapter leads across the Eastern region through biweekly meetings, original training exercises, documenting key insights for management
- Reported progress to senior management through structured weekly presentations; co-organized the National STEM Fellowship conference in collaboration with upper management to successfully execute a national student event

Laboratory Experience at Carleton University February 2025 – April 2025

- Operated oscilloscope, function generator, and multimeter for data acquisition across resistor, capacitor, and diode circuits in AC
- Performed Bode plot analysis and circuit simulation in Multisim, documenting all observations and findings

Carleton Rocketry — Recovery Subsystem December 2024 – April 2025

- Analyzed nylon material variants (Ripstop, Silnylon, Cordura) to determine the best material for parachute canopy, selecting Ripstop for its balance of cost effectiveness, ease of fabrication and tear strength
- Evaluated how panel shape, parachute gore geometry, and size affect drag and stitch integrity from opening shock

EDUCATION

University of Waterloo – Waterloo, ON

Bachelor of Applied Science, Honours Mechanical Engineering | Co-operative Program

September 2025 – Ongoing